أكاديمية طويق
Tuwaiq Academy

1 Introduction

- Goal: The aim of this project is to analyze the Ames Housing dataset and identify the key features that influence house prices.
- Dataset: The dataset contains information about various house attributes such as size, quality, and age in Ames, Iowa.

2 Data Cleaning Summary (Phase 1)

- What you did: I handled missing values, removed outliers (houses larger than 4000 sq ft), and ensured data consistency.
- Why: Cleaning was essential to ensure the accuracy of the final analysis.

3 Feature Engineering Summary (Phase 2)

- Key changes: I created new features like House_Age and price_per_sqft. I also applied Log Transformation to the SalePrice to make its distribution symmetric.
- Encoding: I used One-Hot Encoding for categorical data and Ordinal Encoding for quality features.

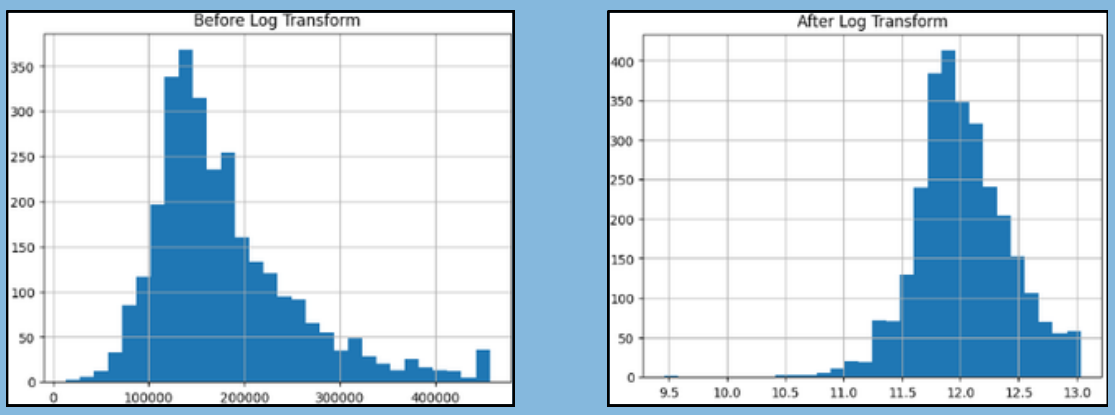
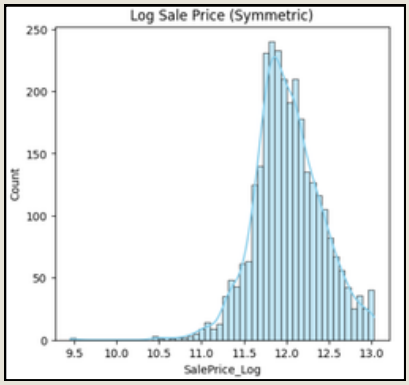


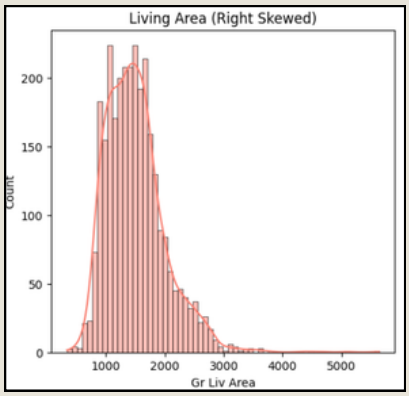
Figure 1: Comparison of Sale Price distribution before and after Log Transformation. The transformation corrected the right skewness, resulting in a normal distribution which is better for analysis.

4

Distribution Analysis

Before analyzing correlations, I examined the distribution of key variables. The target variable (Sale Price) was initially right-skewed, but after applying Log Transformation, it achieved a symmetric normal distribution, which is ideal for statistical analysis.





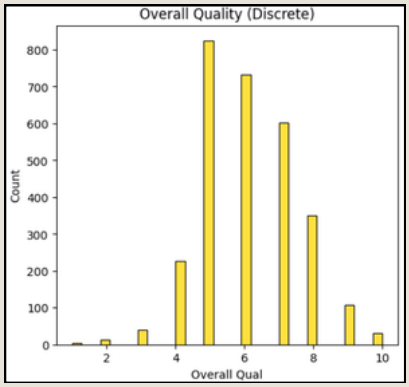
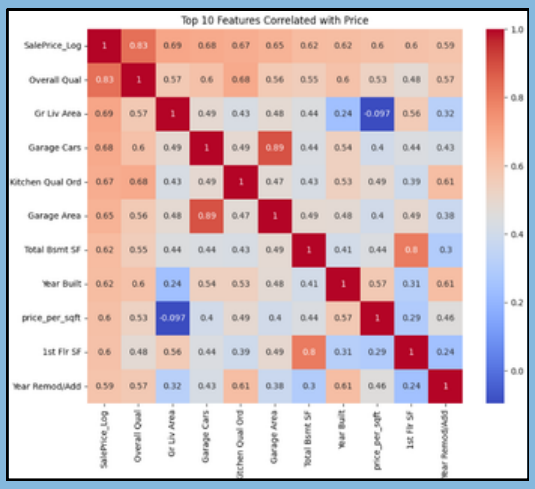


Figure 1: Distribution of Sale Price, Living Area, and Overall Quality. Log transformation helped normalize the price data.

5

Key Findings & Visualisations (Phase 3)

- Discovery 1: Based on the Heatmap, Overall Quality and Living Area have the strongest correlation with house prices.
- Discovery 2: The Scatter Plot shows that newer houses tend to be more expensive even if they have the same size as older ones.
- Visuals:



heatmap
top 10 features
correlated with
price

Figure 2: Correlation Heatmap showing the top 10 features related to house prices. Overall Quality and Gr Liv Area show the highest positive correlation with the target variable.

Scatter Plot
area vs price by
house age
group

Figure 3: Relationship between Living Area and Sale Price, colored by House Age Group. It is evident that newer houses (Blue) generally fetch higher prices compared to older ones for the same square footage.

6

Conclusion

Summary of Findings:
In conclusion, this project successfully transformed a raw, messy dataset into a structured format for analysis. The key takeaway is that Overall Quality and Living Area are the most critical factors determining house prices in Ames. Additionally, the analysis showed that while size matters, the Age of the House plays a significant role, with newer homes maintaining higher value even at smaller sizes.

What I Learned:
Through this project, I gained practical experience in handling missing data, performing log transformations to normalize target variables, and using NumPy for manual statistical calculations. I also learned how to use visualizations like Heatmaps and Scatter Plots to tell a story with data.

Future Steps:
If I had more time or more data, I would:

1. Investigate the impact of specific locations (Neighborhoods) more deeply using geographical mapping.
2. Build a Machine Learning regression model to predict the exact price of a house based on the features I engineered.
3. Analyze the impact of 'Sale Condition' to see if distressed sales significantly lower the market average.